

PROJECT AND TEAM INFORMATION

Project Title

Rapid Aid - A Disaster Response system

Student / Team Information

Team Name: Team #	NanoCore DSCPP-III- 2026-T161
Team member 1 (Team Lead) <i>(Last Name, name: student ID: email, picture):</i>	Sharma, Prajjwal – 2510011461 2510011461@geu.ac.in 
Team member 2 <i>(Last Name, name: student ID: email, picture):</i>	Rawat, Avishkar – 2510010506 2510010506@geu.ac.in 
Team member 3 <i>(Last Name, name: student ID: email, picture):</i>	Dhairya – 2510012143 2510012143@geu.ac.in 
Team member 4 <i>(Last Name, name: student ID: email, picture):</i>	Pundir, Ayush – 2510370008 2510370008@geu.ac.in 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Incidents in smaller environments such as college campuses, residential colonies and clinics are usually managed through telephones, software applications and manual records. Coordinating multiple response units becomes very difficult in smaller localities and when there are multiple incidents.

The reason for the development of this project is automation of this management and coordination process. This is done based on simple factors like the severity of the incidents, waiting times(times of the incidents that have not been acknowledged), number of people affected by the incident, availability of units and the distance of the available unit from the location.

Computer aided dispatch systems are used usually in larger scale emergencies and disasters, but they need huge infrastructure and are designed specifically for large scale organisations.

Rapid aid is an implementation of similar computer aided dispatch system on a smaller scale for smaller localities.

Project Goals and Milestones (2 pts)

The goal of this project is design and implement a Disaster Response software that can simplify the allocation of resources and response units at disaster struck locations. This is designed mainly to be implemented in a small local area where larger computer aided dispatch systems are difficult to implement.

The milestones of this project are:

- Implementing an incident prioritisation system. This prioritisation, will depend on the severity of the incident, number of people involved in it or affected by it, availability of relevant response units and resources at that moment and other factors.*
- Representing the local area using weighted graphs. Adding methods for adding new maps as graphs and add new nodes in existing graphs.*
- Implementing smart allocation of resources like hospitals, shelters, food and water supplies and emergency vehicles, based on priority of the incident.*
- Implementing method of collaboration between different emergency departments for a individual disasters.*
- Storing the records of past allocations, incidents and reports for future reference. This data will be stored on the device of the dispatcher, though this can be connected to cloud for longevity of data.*
- Presenting a statistical overview of past incidents and successful/unsuccessful allocations and dispatches from stored logs.*
- Implementing a simple console based UI for the dispatcher. This ui will provide an interface to view and allocate resources, view past reports, enter new reports, check report status.*
- Allowing flagging of fraud or false reports, prank calls by the dispatcher.*

State of the Art / Current solution (1 pt)

Several CAD systems and technologies have been developed in different countries to provide fast response to disasters or emergency incidents. Examples of this type of technology in use are ERSS (Emergency response support system)-112 in India, NG-112 in the European countries, etc. The NG-112 has not yet been deployed fully, which shows one of the major reasons of the difficulty of using this technology. NG-112 runs using IPs but switching the current emergency response infrastructures to IP is taking a lot of effort, time, large scale public funding and a lack of strict mandates, leading to long delays in the full scale deployment of this service in Europe.

Moreover, these services are designed for deployment at a large scale, and are often overseen by big and government associations like the home ministry of India. In the case of smaller associations and localities, emergency services are still called through telephone and messaging, and the reports are handled manually. Poorly managed incident reports and the lack of readily available emergency services in the The inclusion of fraudulent or prank calls and false alarms and the lack of nearby emergency services, leads to difficulty in the use of such software in smaller communities and regions like cities, by universities, in workplaces (industrial or others).

Project Approach (3 pts)

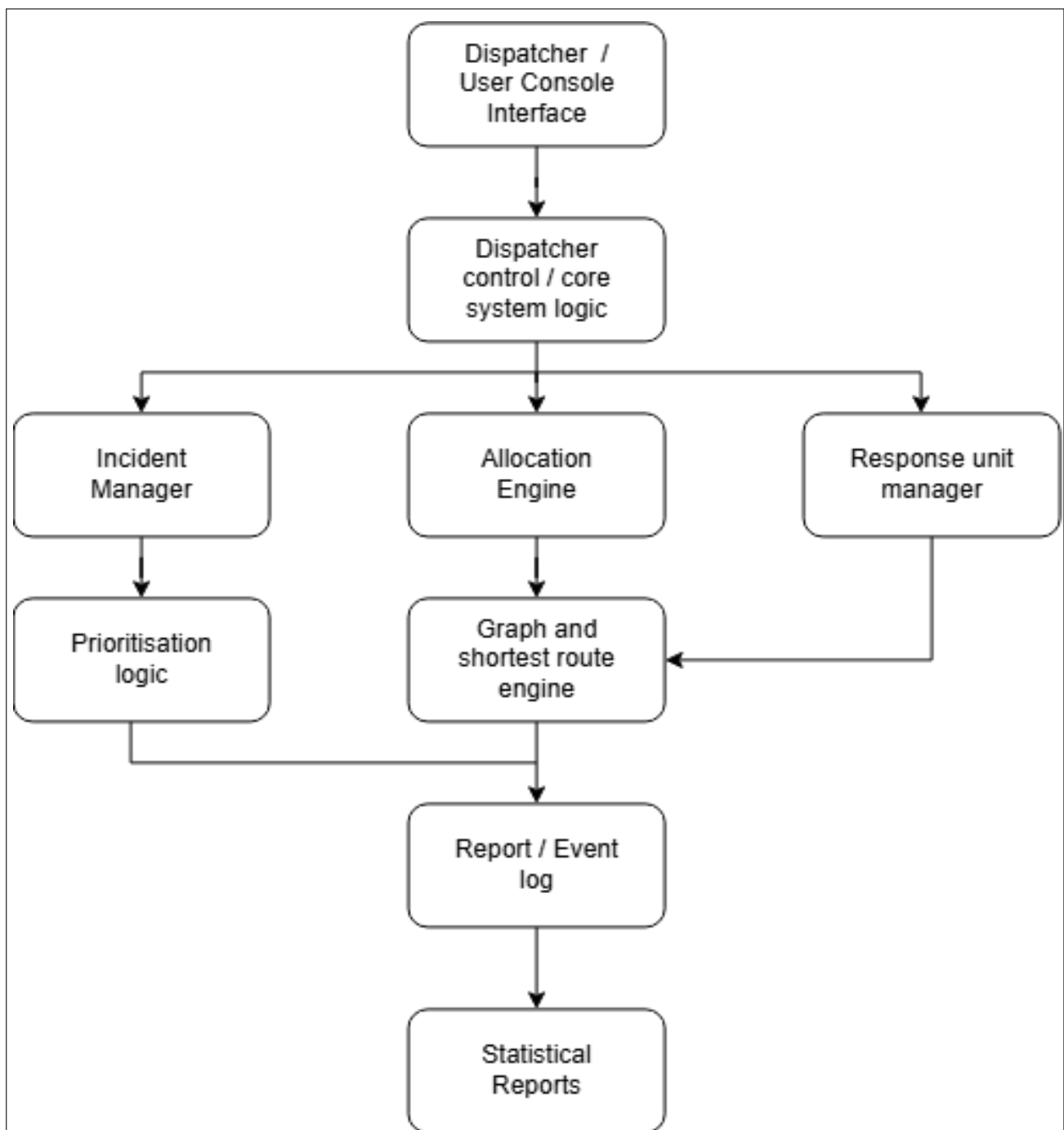
This software will be designed to track the assignment and availability of the required resources. It should assign them to the incidents depending on the distance between the service and the destination of the incident. The user should be able to input incoming incident reports and the software should use this data to allocate the resources. The user should be able to review the currently allocated resources as well as status of current reports. The user should also be able to add new maps as graphs or new locations to the existing map. Also, blocking of paths and traffic (input by the user or through the report) should affect the shortest route logic.

This will be done through the following steps:

- The surrounding locality has to be mapped and a rough idea of the area, major buildings, emergency service locations and distances between them should be taken to properly represent the area using a weighted graph data structure.
- A system must be developed which should calculate and find the shortest path between two locations. This system should be adaptive to factors like traffic, crowded paths, blocked roads, etc.
- A system will be developed which should accept the reports of incidents input by the user and determine the priority of the incident, depending on severity of incident and availability of the required resources at the time.
- The system should be able to prioritise between different incidents using the data provided by the report. Higher severity incidents should be dealt with before lower priority ones whenever possible.
- The quality of the help provided should not be reduced. The already allocated resources from lower priority incidents should never be reassigned to a different higher priority incident, unless necessary.
- The system must be able to smartly allocate these resources.
- The system must calculate the statistics of past successful/unsuccessful dispatches when requested by the user.

The system will not aim to automate the process of dispatching. The software will still depend on human diligence, judgement and intelligence for dispatch, it will only simplify the process.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pt)

The expected outcome will be a complete software that will provide a simple Disaster Response system for adaptive dispatching in local communities. It will provide a simple UI for the dispatcher to coordinate the resources, check past reports, add new maps as graphs or add new nodes(locations to existing graphs). It will provide an on device logging to store the incident reports and dispatch records for future use and statistical reports. The final outcome will be able to coordinate multiple response systems for any incident. It will be able to handle different severities of incidents without too much manual effort from dispatcher end. It should also be able to allocate resources such as hospital rooms, shelters, food and water supply, etc. in disaster struck areas. It should also be scalable. Later, AI-based NLP detectors can be used to enable the service to ignore false alarms and unnecessary or fraud reports to reduce the wastage of time and improve the response quality and time.

References

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